Multimodal Alpha Forecasting and Robust Portfolio Optimization

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1 Introduction

2 Data

2.1 Data Sources

Because no single source provides all the information required for forecasting and portfolio construction, we integrate data from multiple independent providers. Our final dataset combines four major categories of information:

- **S&P 500 Constituents:** The list of S&P 500 tickers during our sample period serves as the base universe.
- **Stock Market Data:** Daily OHLCV (Open, High, Low, Close, Volume) data for all S&P 500 stocks from 2016–2023 are obtained from Yahoo Finance.
- **Company Fundamentals:** Firm-level financial metrics (e.g., sales growth, margins, ROA, ROE, leverage, liquidity ratios, valuation ratios) are sourced from WRDS/Compustat.
- **Macroeconomic Indicators:** Broader economic conditions are captured using FRED variables such as GDP growth, unemployment rates, and Treasury yields.
- **News Data:** Daily financial news headlines and summaries are collected from the FNSPID dataset.[1]

2.2 Data Processing Pipeline

After gathering raw inputs, we construct a unified panel dataset at the (ticker, date) level. The main steps are:

1. **Cleaning and Alignment:** All data sources are standardized to daily frequency. We merge by ticker and date, forward-fill fundamentals to match market trading days, and remove observations with invalid or missing tickers.
2. **Return Construction:** Using daily closing prices, we compute the next-day stock return, which serves as the prediction target.
3. **Feature Engineering:** We generate:
    - market-based indicators (returns, volatility, volume changes),
    - fundamental ratios (profitability, leverage, valuation),
    - macroeconomic variables aligned to each trading day,
    - sentiment scores from FinBERT applied to news headlines,

[1] https://huggingface.co/datasets/Zihan1004/FNSPID

- PCA embeddings from news text for richer semantic representation.

4. **Handling Missing Values:**
   - *Forward-filling time-invariant or slowly changing variables* (e.g., fundamentals such as leverage, ROA, and margins). These variables are forward-filled *within each firm* so that only past information is used, which prevents look-ahead bias.
   - *Zero-filling structurally missing ratios or indicators* (e.g., variables undefined for certain firms or years). This approach preserves the panel structure without imputing information from the future.

5. **Train–Validation–Test Split:** To avoid look-ahead bias, we split the dataset chronologically:
   - 2016–2021 for training,
   - 2022 for validation,
   - 2023 for out-of-sample testing.

This pipeline produces a complete, aligned dataset suitable for both prediction modeling and portfolio backtesting.

3 Methodology

3.1 Stage 0 - Multimodality

- Tabular data
- Text data

3.2 Stage 1 - Stock Next Day Return Prediction

3.3 Stage 2 - Robust Portfolio Optimization

4 Discussion

5 Future Work

6 Conclusion

7 Appendix